

NATIONAL UNIVERSITY OF SINGAPORE
Department of Mathematics

2026-2027 (Semester 1)

MA1513

Lecture Problem Set 1

Instructions

- To be discussed during lecture in week 1 (Aug 11 and 13, 2026).
 - Note that you do not need to submit the answers to this problem set.
 - To prepare for the problem set, students should view the "Week 1 key video" before the lecture.
 - Students who would like to try out the problems in advance may refer to the hints given at the end of the problem set.
 - You need to install MATLAB software and use the commands for some of the questions. (Refer to the PDF document: MATLAB for Engineering for details)
1. Solve each of the following systems by Gaussian Elimination or Gauss-Jordan Elimination.

$$(a) \begin{cases} x_1 + 3x_2 + 3x_3 + 2x_4 = 1 \\ 2x_1 + 6x_2 + 9x_3 + 5x_4 = 5 \\ -x_1 - 3x_2 + 3x_3 = 5 \end{cases} \quad (b) \begin{cases} x + y + 2z = 4 \\ x - y - z = -1 \\ 2x - 4y - 5z = 1 \end{cases}$$

$$(c) \begin{cases} u - v + 2w = 6 \\ 2u + 2v - 5w = 3 \\ 2u + 5v + w = 9 \end{cases}$$

2. Given a quadric surface in the xyz -space with equation

$$ax^2 + by^2 + cz^2 = d,$$

where a, b, c, d are real constants, that passes through the points $(1, 1, -1)$, $(1, 3, 3)$ and $(-2, 0, 2)$, find a formula for the quadric surface.

3. Consider the population of certain endangered species of wild animals: On the average, each adult will give birth to one baby each year; 50% of the new born babies will survive the first year; 60% of the one-year-old cubs will survive the second year and become adults; and 70% of the adults will survive each year.

Let x_0, y_0 and z_0 be the numbers of babies, one-year-old cubs and adults, respectively, at the end of 2020, and x_1, y_1 and z_1 be the respective numbers at the end of 2021.

- (a) Based on the information given above, find a 3×3 matrix $\mathbf{A}$ such that

$$\begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} = \mathbf{A} \begin{pmatrix} x_0 \\ y_0 \\ z_0 \end{pmatrix}.$$

- (b) Let $\begin{pmatrix} x_n \\ y_n \\ z_n \end{pmatrix} = \mathbf{A}^n \begin{pmatrix} x_0 \\ y_0 \\ z_0 \end{pmatrix}$ where n is a positive integer. What do the numbers x_n , y_n and z_n represent?
- (c) (**MATLAB**) Suppose initially, $x_0 = 0$, $y_0 = 0$ and $z_0 = 100$. What will be the populations (i) three years later; (ii) ten years later? (See Hints for some help with MATLAB commands.)

4. Let $\mathbf{A}$ be a 3×4 matrix. Suppose that

$$\begin{cases} x_1 = 1 \\ x_2 = 0 \\ x_3 = -1 \\ x_4 = 0 \end{cases}$$

is a solution to a non-homogeneous linear system $\mathbf{Ax} = \mathbf{b}$ and that the homogeneous system $\mathbf{Ax} = \mathbf{0}$ has a general solution

$$\begin{cases} x_1 = t - 2s \\ x_2 = s + t \\ x_3 = s \\ x_4 = t \end{cases}$$

where s, t are arbitrary parameters.

- (a) Find a general solution for the system $\mathbf{Ax} = \mathbf{b}$.
- (b) Write down the reduced row-echelon form of $\mathbf{A}$.
- (c) Do we have enough information to find the matrix $\mathbf{A}$?

Hints and answers

1. (a) General Solution: $x_1 = -2 - 3s - t$, $x_2 = s$, $x_3 = 1 - \frac{t}{3}$, $x_4 = t$.
(b) is inconsistent. (c) Unique solution.
Refer to video clips 1.2 and 1.3 if you need help with solving the linear systems.
2. If the surface passes through a certain point, the coordinates (x, y, z) of the point will satisfy the equation of the curve. You should set up a homogeneous system with four variables.
3. First set up a linear equation by expression x_1 in terms of x_0, y_0, z_0 . Take note of the contributions of the babies (x_0), one-year-old cubs (y_0) and adults (z_0) in 2020 to the number of babies (x_1) in 2021. Do the same for y_1 and z_1 . Then convert the three equations into the required matrix equation form.
(a) $\mathbf{A} = \begin{pmatrix} 0 & 0 & 1 \\ 0.5 & 0 & 0 \\ 0 & 0.6 & 0.7 \end{pmatrix}$;
(c) (i) Approx. 49, 35, 64
MATLAB commands:
 - Key in matrix $\mathbf{A}$: $\gg \mathbf{A} = [0 \ 0 \ 1 ; 0.5 \ 0 \ 0 ; 0 \ 0.6 \ 0.7]$
 - Key in the initial population as a column matrix: $\gg \mathbf{C} = [0 ; 0 ; 100]$
 - Key in a matrix power, say $\mathbf{A}^3$: $\gg \mathbf{A}^3$
 - Key in a matrix multiplication, say $\mathbf{AC}$: $\gg \mathbf{A} * \mathbf{C}$
4. (a) Recall that the general solution of a non-homogeneous system is related to the general solution of its associated homogeneous system.
(b) $\begin{pmatrix} 1 & 0 & 2 & -1 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$. Recover this RREF of $\mathbf{A}$ from the general solution of $\mathbf{Ax} = \mathbf{0}$. Refer to chapter 1 note, last part of section 1.3.